

Physical Manifestation of Generative AI Music Systems for Live Performance

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Figure 1: Jordan Rudess performing with the JAM_BOT at the MIT Media Lab. Credit: *Jordan and the JAM_BOT: A Work-in-Progress Performance*, 2024. Photo: Caroline Alden. Courtesy of MIT.

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Abstract

This paper explores the physical manifestation of generative AI music systems for live performance, focusing on bridging the expressive gap between AI-generated music and audience perception. Through a year-long collaboration with a human performer, we constructed a kinetic sculpture that visualizes the outputs of an AI JAM_BOT during concerts. The sculpture, powered by ML-based and pattern-driven mapping methodologies, interprets real-time AI musical decisions as expressive movements. Audience feedback indicates increased engagement and curiosity, although interpretability remains a challenge. Our work highlights the potential of embodied

visualization to establish communicative presence for AI performers and suggests avenues for future research.

CCS Concepts

• **Applied computing** → **Performing arts**; **Sound and music computing**; • **Human-centered computing** → *Collaborative interaction*; *Interaction devices*.

Keywords

Live, Performance, Music, Visualization, Sculpture, Kinetic Sculpture, Artificial Intelligence, Generative AI, Audience, Embodiment, Interaction, Perception

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1 Introduction

In previous work, we introduced the concept of JAM_BOTS, real-time generative AI-powered systems for musical accompaniment, and demonstrated how they can augment live music performances [Blanchard et al. 2025]. As we continue our research into these novel AI-powered systems, we observe an emerging need for communicative visualization. Although a generative AI music system does not require a physical manifestation in order to produce music and accompany human musicians onstage, expressive visualization is essential to bridge the gap between the music and the AI performer from the audience’s perspective. For human performers, this gap is naturally filled through human expression and the visible performance of an instrument (keyboard, voice, violin, etc.). These visual cues help audiences understand which performer is producing which sound [Saldana and Rosenblum 1993]. However, audiences unfamiliar with AI performers may lack intuitive expectations for how such agents should appear or behave onstage. This simultaneous demand for visibility and the absence of conventional audience expectations has inspired a new domain of artistic inquiry: *Manifestation and visualization of the AI system*.

On September 21, 2024, we concluded a year-long collaboration with GRAMMY-winning keyboardist Jordan Rudess, culminating in a sold-out performance at the MIT Media Lab. The performance featured Rudess improvising alongside a custom-trained JAM_BOT on 6 distinct pieces spanning a range of musical styles. To manifest and visualize the generated outputs of the AI system, we constructed a 16-foot-tall kinetic sculpture integrated into the live stage performance. Using a custom-trained, multimodal generative AI model for Music-to-Motion Mapping, along with a series of Pattern-based Mappings, we enabled the AI system to exhibit an expressive and communicative stage presence. In the sections that follow, we detail the design of our system and the mapping strategies used to convey the AI’s presence visually.

The demonstrated causality of computer-based musical instruments is not a new issue. Visuals that closely link to the musical or audio component of a performance extend the instrument to make it more understandable and allow the audience to interpret

the performance as a live production [Paradiso 2004]. In some cases, computer-based sounds may be communicated in the form of humanoid virtual characters. Crypton Future Media’s “Hatsune Miku” is a voicebank for the Vocaloid voice synthesis software [Yamaha Corporation 2024], but the voicebank is packaged and marketed with graphics and narratives that allow Miku to build a following [Crypton Future Media [n. d.]]. Other systems intertwine the synthesis of audio and visual components into one cohesive interactive application. Golan Levin’s “Yellowtail” (1998) and related pieces allow a user to animate visual artworks using mouse movements [Levin 2000b]. As demonstrated in the audiovisual performance of “Scribble” using Levin’s Audiovisual Environment Suite (AVES), the animated elements of each interactive visualization produce sounds that map to their motion and appearance [Levin 2000a].

2 System Overview

2.1 Human and AI Performers

Generative AI agents for live music performances such as JAM_BOTS generate long-term musical structures using a Music Transformer [Huang et al. 2018] every time they are prompted. The prompting of these agents occurs according to various interaction modalities with the human performer playing notes on a keyboard, which dictate when and how their contexts should be managed. Unlike human performers, their predictions of where a live improvisation might go in the next few bars can easily be extracted. We call these potential unrealized musical trajectories “*musical futures*”. As generative AI increasingly finds its way onto the live stage, we believe that there is a compelling opportunity to design systems that can not only generate these *musical futures*, but also visualize them—both how they are realized and how they are adjusted—giving audiences insight into how AI responds to its human counterparts.

2.2 Kinetic Sculpture

For this performance, we represent the AI music generation system through kinetic sculpture. We constructed a 5 x 5 grid of novel motion-control fixtures (known as the Generic Pan Tilt) featuring leaves mounted on infinite-pan and 180-degree-tilt axes [Naseck et al. 2025]. When operated together, the system of fixtures may display emergent patterns, expressive sweeps, and subtle movements. The leaf geometry was chosen since the motion patterns can replicate both slow, flowing movements and fast, sudden movements of leaves blowing in the wind. The structure, interacting with the performer onstage, served as the visual manifestation of the expressive intent of the AI system, as can be seen in Figure 1.

The system is designed to fully leverage the audience’s lack of prior expectations. Unlike the familiar lighting or media systems used in modern performance venues, the sculpture’s presence onstage sparks curiosity even before the performance begins—much like an elaborate set design at a theatrical production. Though initially mysterious, the sculpture integrates seamlessly into the stage environment, achieving the “*visual fit and focus*” needed to feel natural rather than imposed [Blanchard et al. 2024]. Importantly, the audience does not immediately understand the sculpture’s kinetic capabilities or how they relate to the music. As a result, the mapping between sound and movement can take on a more abstract,

expressive form, akin to human dance. Just as a dancer interprets music through motion, the sculpture interprets the AI-generated music, and in turn, audiences interpret the sculpture’s performance.

The kinetic sculpture enables audiences to learn and interpret the AI system’s expressive output over time, as they would with a human performer. This fosters recognition of the sculpture’s unique mappings, as well as differentiation between the AI’s musical output and that of human collaborators. In compositions where performers use abstracted or synthetic instruments (e.g., playing a marimba sound from a keyboard interface), time-aligned bodily movements are key indicators of a performer’s specific musical contribution [Thompson et al. 2005]. A violinist may bow more forcefully to increase volume or a pianist may move expressively across octaves; these visual cues provide *musical transparency* [Blanchard et al. 2024]. Providing this kind of transparency for AI systems is more complex. Traditional visualizations—such as sheet music or piano rolls—require interpretive literacy and cannot be directly perceived as expressive. Direct visual mappings, such as a light that pulses when a note is played, similarly lack expressivity. The sculpture’s wide range of motion and emergent behavior supports the development of a consistent visual language over time. This consistency helps audiences form expectations and intuitively “read” shifts in the AI’s musical phrasing—just as one might anticipate a breath before a vocalist hits a high note.

3 Methods

To bridge the perceptual gap between generative music and its onstage manifestation, we explored two mapping methodologies. These methods animate the kinetic sculpture with expressive movements that reflect the musical outputs of the JAM_BOT in communicative and performative ways.

Each motion control fixture in the sculpture uses DMX512-A [Entertainment Services and Technology Association 2024] as its control protocol. Pan and tilt values must be computed in real time by choreographing and rendering the sculpture’s movements. An application written in Processing [Processing Foundation 2024] consumes Open Sound Control (OSC) inputs from the JAM_BOT and a separate ML-based Mapping. These inputs are then passed to the system’s Pattern-based Mappings for fixture parameter computation. A screenshot of the application performing a flower-opening choreography can be seen in Figure 2 and the final motion output can additionally be seen in the video (Appendix A). A system diagram can be seen in Figure 3.

In this section, we describe our ML-based Mapping methodology, followed by our Pattern-based Mapping.

3.1 ML-based Mapping

Jordan Rudess is particularly famous for the virtuosic, expressive, and fast-paced melodies that he performs while improvising. One of the modalities of the JAM_BOT is the modeling of this exact type of melodies, conditioned on a sequence of chords played live by Rudess. To visualize these musical gestures, we developed a real-time machine learning system that translates MIDI-based musical input to expressive and abstract hand-like motions. These movements, interpreted through 3D coordinates, drive the kinetic sculpture’s motion patterns and serve as a visual proxy for the AI’s musical

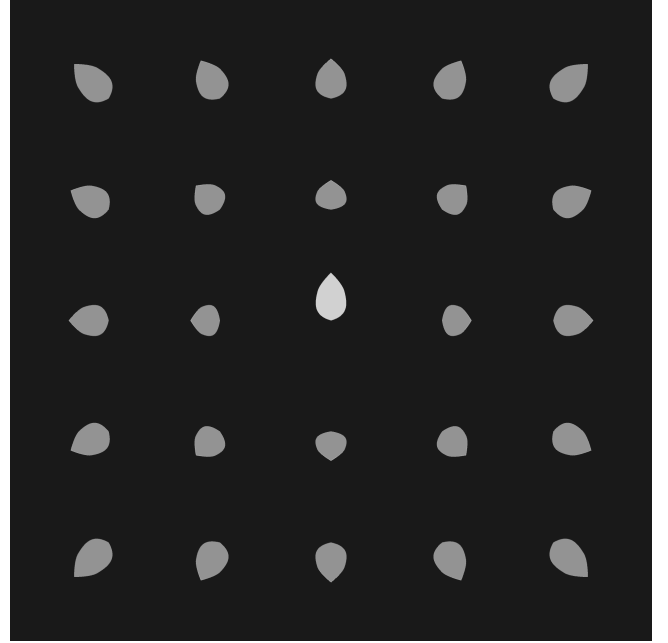


Figure 2: Screenshot of control application performing a flower choreography

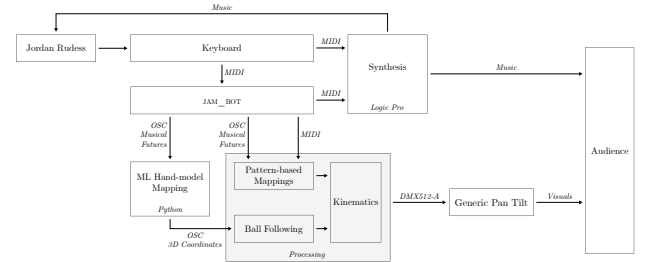


Figure 3: System diagram for the performance

intent. This allows us to create an implicit mapping, alleviating the need for designers to create arbitrary and explicit mappings (such as pitch, note onset, etc.).

This mapping system builds on FACT [Li et al. 2021], a Transformer-based Neural Network originally developed by Google to generate human dance motion from musical input. We used this architecture to develop our own bimodal Transformer-based neural network capable of generating *hand motions* based on a MIDI musical input. We trained our model on a custom dataset of “hand-dancing” gestures, captured via motion-tracked individuals improvising expressive hand motions in response to diverse musical excerpts. Critically, this dataset avoids rigid parametric mappings (e.g., pitch-to-color), instead privileging a perceptual, interpretive connection akin to how human dancers embody music. This allows the model to learn musically aligned, temporally coherent movements—including beat-synchronized pulses, chord-driven transitions, and intensity-responsive gestures—that feel intentional without being

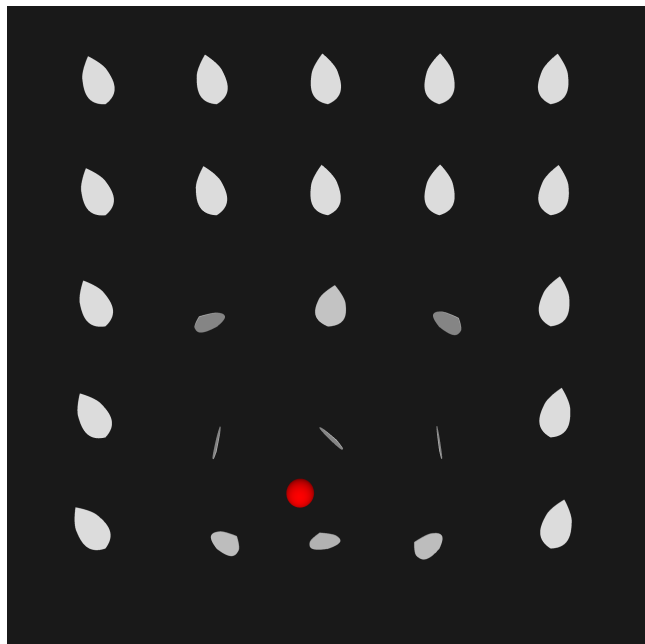


Figure 4: Rendered leaves following virtual ball led by the ML-mapping system

literal. Through this method, the sculpture can achieve “*live improvisational reactivity*” where the system may expressively respond to any musical input [Blanchard et al. 2024]. This idea is reminiscent of Marc Downie’s “Loops” (2005), a “digital portrait” of a motion captured hand performance of dancer and choreographer Merce Cunningham [Downie 2005].

The output of this system is integrated with the sculpture through a 3D coordinate-following system. A virtual ball moves around the virtual space of the sculpture, driven by the low-latency Transformer-based neural network (Figure 4). As this ball moves, the nearest leaves (within a radius of 2–3 leaves) turn to point to it. Large spatial displacements of the ball create sweeping, gestural motions; small displacements produce subtle pulsing, vibrating, or oscillatory leaf movements. This range of expressivity allows the AI mapping to fully engage the kinematic ability of the sculpture without restriction. The tilt function of the fixtures represents the distance between the ball and each fixture, creating a natural gradient of motion as the ball moves through space.

This real-time system embodies the entangled agencies of AI and human artist. It transforms opaque AI outputs into physical gestures that feel performative, enhancing the audience’s ability to attribute musical expression to the AI. This visual mapping is representational, expressive, and reactive, and enables the AI to “move” in response to live changes in the human performer’s input in a way that provokes audience interpretation. In doing so, it not only affirms the AI as a co-performer but also invites the audience into a new visual-musical vocabulary that evolves with every performance.



Figure 5: Jordan Rudess performing with violinist and vocalist Camilla Bäckman as the AI rests. Credit: Jordan and the JAM_BOT: A Work-in-Progress Performance, 2024. Photo: Caroline Alden. Courtesy of MIT.

3.2 Pattern-based Mappings

In addition to the machine learning-based approach, we implement several algorithmic, Pattern-based Mappings. These mappings produce defined, reproducible choreography derived from the AI’s music outputs and planned *musical futures*. Unlike the gestural and improvisatory character of the ML-based Mapping, these mappings are deterministic and rule-driven. These Pattern-based Mappings are controlled by specific indicators in the AI-generated music described below.

3.3 Resting

The sculpture animates only when the AI is actively generating music. Early in the performance, we introduced a call-and-response modality in which the human performer and AI alternated musical phrases. This back-and-forth repeats as the human elaborates or reacts to the AI’s contributions. During the resting phases—when the AI is not producing sound—the sculpture enters a listening state. Its leaves gently sway, signaling that the AI is inactive but attentive (Figure 5). Once the AI plays its first note, the sculpture transitions to an expressive state, with the leaves springing to life to visualize the generated music. The resting state is determined by looking into the AI’s future playback buffer, or *musical futures*. Rather than using a timeout function based on the last note played, the sculpture looks ahead to see if a note is scheduled to play in the future. If no note is scheduled, the sculpture enters the resting state.

This pattern is essential to communicating to the audience that the sculpture reflects only the AI’s musical ideas. This distinction plays a key role in establishing *musical transparency*. Although the performer verbally introduces this mapping near the beginning of our performance, such an announcement cannot be expected of every performance. Therefore, the sculpture must visually communicate this relationship on its own, making its alignment with the AI system intelligible through its presence and motion alone.

3.4 Regeneration

A particularly novel feature of the AI music generation system is its ability to plan ahead. Unlike human performers, whose musical intentions remain internal, the AI's prescribed future can be externally visualized through the sculpture. This offers an unprecedented window into the system's anticipatory logic. As the performer plays and the AI listens, the AI may choose to regenerate its planned musical trajectory. Visualizing this moment is critical, as it symbolizes a shift in the AI's musical direction—a change not just in output, but in intent.

Throughout the performance, these regeneration events are signaled by swift, 360-degree rotations of the sculpture's leaves. In collaborative modalities where the human and AI perform simultaneously, these motions naturally accompany moments of transition, as the AI reorients its musical thought mid-stream. In contrast, during non-simultaneous modalities—where human and AI alternate musical sections—the AI uses this gesture at the start of each segment it initiates. This rotation serves as a visual cue that it has completed its internal reasoning and is about to perform its next musical idea.

4 Audience Evaluation

We conducted a survey to determine whether audience members noticed and interpreted the interaction between the AI-generated music and the movements of the kinetic sculpture. Of the 41 audience members who consented to participation, five were excluded for lack of completion of any portion of the survey, resulting in 36 responses. As part of this survey, we included a series of perceptual checks of five features of the sculpture, some of which were genuinely present in the performance and others which were intentionally fabricated, to assess whether participants were actively noticing and accurately reporting the behaviors of the sculpture:

- (1) Whether the leaves moved in time with the accompaniment (Present)
- (2) Whether the leaves moved rapidly as the songs switched directions (Present)
- (3) Whether the leaves correlated to the sound of the keyboard (Red-Herring)
- (4) Whether the leaves correlated to the sound of the violin (Red-Herring)
- (5) Whether the leaves correlated to the sound of the bassline (Red-Herring)

Chi-square goodness-of-fit tests indicated that participants were significantly more likely to report noticing Feature 2 ($\chi^2(1, N = 28) = 14.29, p < 0.001$) and significantly more likely to report not noticing Feature 5 ($\chi^2(1, N = 28) = 7, p = 0.008$). Several open-ended responses reflected engagement with the sculpture's kinetic behavior, especially the motion of the leaves. One participant wrote, "I saw the leaves turning in a pattern," while another noted, "Even the leaf that was in trouble was great... it was a kind of mechanical improvisation!" Others participants indicated that they were engaged and reflective about the sculpture's behavior, even when the mappings were not fully clear. One noted, "Wish we had more of a key to what the sculpture was doing to those who have less of a musical background and ear for subtle changes... but it was still a great visual aid and generally cute and fun to watch."

Another reflected, "The kinetic sculpture was useful as an indicator of AI activity, but I wish I'd had a model for why it did what it was doing." These comments reinforce the idea that while the sculpture drew meaningful attention, its interpretability varied across the audience.

5 Future Work and Discussion

5.1 ML-based Mapping

In future work, we aim to expand the motion vocabulary beyond hand gestures to include full-body or style-conditioned movement. This would be achieved by applying the same methods using a dataset of full-body dance instead of hand-only motion. This would enable richer and more diverse visual expressions of AI-generated music, better capturing the nuances of different musical styles and moods, as these factors are more visible in human dance than hand-only motion. We are also interested in incorporating real-time adaptability, allowing human performers to influence the mapping dynamically during performance. We may apply motion capture to the live performance itself in addition to the training process. These additional live capture parameters may map to translations and transformations on top of the ML-based model, affecting the scale and positioning of the output motion. These directions continue our exploration of AI as an expressive, performative collaborator—one whose visual presence evolves in tandem with its musical contributions.

5.2 Pattern-based mapping

As noted by audience members, the sculpture communicates itself as a representation of the AI but fails to consistently demonstrate in depth how the AI is composing music. We hope to explore deeper alignment between the sculpture's movements and the AI's musical reasoning through in-depth musical analysis—both of the pre-buffered *musical futures* and the actively played notes. Rather than relying solely on time-based transitions (e.g., playing vs. resting, or regeneration events), we propose analyzing structural elements of the music—such as motifs, parallelism, energy, and tension. These features could drive new, more intelligible mappings. For instance, repeated motifs could trigger recurring animations, allowing the sculpture to develop a kind of "visual musical vocabulary." As the AI alters or extends these motifs, the corresponding animations could also evolve in real-time. This more musically aware system may improve interpretability and bring the sculpture closer to "*demonstrating virtuosity*", where the complexity and rigor of the AI's musical ideas are reflected in its movement [Blanchard et al. 2024]. In doing so, the sculpture could not only embody the AI's output but also communicate the depth of its musical structure.

5.3 Kinematic System

Currently, each pattern-based animation (such as the flower animation) is manually and individually programmed prior to each performance. We would like to explore how a kinematic solver with a graphical user interface may be used to map sculpture movement to the AI-generated music in real-time. Similar to the role of a lighting or media designer at a live concert using purpose-built consoles, the operator may draw paths, shapes, and other inbetweening tasks found in computer graphics animation. These functions may be

specifically adapted for use with live kinetic sculpture. By combining this approach with the aforementioned automated motif analysis, we envision a system where designers can visually author expressive movement in response to the structure of AI-generated music, enabling more nuanced, responsive, and artistically controllable performances.

5.4 Performer Communication

While the current system provides abstract visualization for the audience, it does not provide direct and rigid communicative utility to the other performers on stage. Specifically, it lacks “*communication for collaboration*”—the intentional signaling that enables real-time musical interaction between performers [Blanchard et al. 2024]. Future iterations of the system should incorporate more explicit, identifiable outputs or an additional separate visualization that allows performers to directly anticipate and interpret the AI’s musical direction. This type of visualization may take the place of head movements and other gestures that human performers use to signal to each other [Williamon and Davidson 2002]. Enabling this kind of inter-performer signaling would strengthen the AI’s role as a collaborative performer.

6 Ethical considerations

As with any artificial intelligence music and visual system, the introduction of generative tools raises valid concerns about the potential displacement of human artists and designers. Our aim is not to replace creative professionals but to augment the ability of performers and production staff to allow them to create new experiences. In particular, this visualization tool allows designers to interpret and anticipate how the AI music generation system produces music, but it does not provide an end-to-end solution of music-to-choreography. Each mapping to the sculpture is explicitly and deliberately defined and designed by a human operator; the system only provides more inputs, similar to the function of a tap tempo or noise function on a lighting or sound console. By foregrounding human authorship and control, we position the system as a creative aid: a tool that expands, rather than replaces, the expressive possibilities available to live performance productions.

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A Video

A compilation of demonstrative clips from the performance can be found in the supplemental materials for this paper.